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2.
 - a.
 - 1) The robot must resemble a human closely enough that it doesn't fall into the "Uncanny Valley" as that can significantly impact how visitors view the robot and the exhibition as a whole.
 - 2) The robot must navigate through crowds of people and through different rooms without interrupting interactions between different people, careful not to obstruct exhibitions or even crash into it. It also has to maintain proper distance between each visitor to not violate any personal boundaries.
 - b. They need to be able to physically adapt to different people, especially those with limited mobility. For example, helping a child put their clothes on is far different from helping someone with say, a healing spine injury, put their clothes on. The robot needs adaptive motor skills as well as levels of forcefulness in their actions that's quite difficult to pull off.
 - c. The delay between someone giving commands and the robot actually pulling it off in time can be very crucial and can lead to potentially very harmful mismatches. For example, if someone is undergoing a surgery and a robot is commanded to make an incision, due to the delay it could lead to the wrong tissue/organ being cut which could have disastrous consequences.
3. The robot should be skilled at real-time interaction recognition. For example, it should be able to decipher what certain gestures mean in certain contexts and be able to guess what the task is based on the clues. If a chef were looking frantically for an object while their food is frying and they reach out a hand towards you, they probably expect a spatula or a stirring tool. Additionally, robots need to be able to swiftly and adeptly maneuver around all the chefs in the kitchen, careful not to bump into anyone carrying dishes or food/obstruct anyone's path. They should also be able to maneuver around in close quarters to avoid potentially spilling stuff on other chefs or having too close of workstations. One final concern could also be the "Uncanny Valley" aspect: if the robot were to also be serving food to the customers it needs to be designed in a way that visitors view the robot as friendly rather than it having fallen into the "Uncanny Valley" category.

To do this, you should first and foremost consult the kitchen staff as they will be the ones mostly interacting with the robot. Their workflow and communication style needs to be properly understood for the robot to be properly integrated into the kitchen environment. Additionally if the robot were also to be serving food, customers also need to be

consulted to judge their opinion of the robot. Finally HRI professionals and designers should also have a say as they are designing the final product.

4. One novel use case that I can think of could involve being a robotic, emotional support companion at home. If the robot senses that a user is sad it could tilt its head down and ask you what is wrong, as well as suggesting things that could make the user feel better such as good ice cream places in the area. If the user is in a good mood the robot can be alert and happy when the user walks in the room to show that it appreciates the user coming back. This can help the user connect with the robot almost like if it's a pet dog through a sense of personality and human-ness making the person who is interacting with the robot more likely to develop a bond and share details an otherwise inanimate robot wouldn't (like Alexa or Google Home). This can also let people get the help they need if they are going through a rough time, such as connecting someone with a therapist or psychiatrist.
5. Attached below:

